A Countable-Tail Obstruction to the $C(K)$ Cube-Not-Square Route

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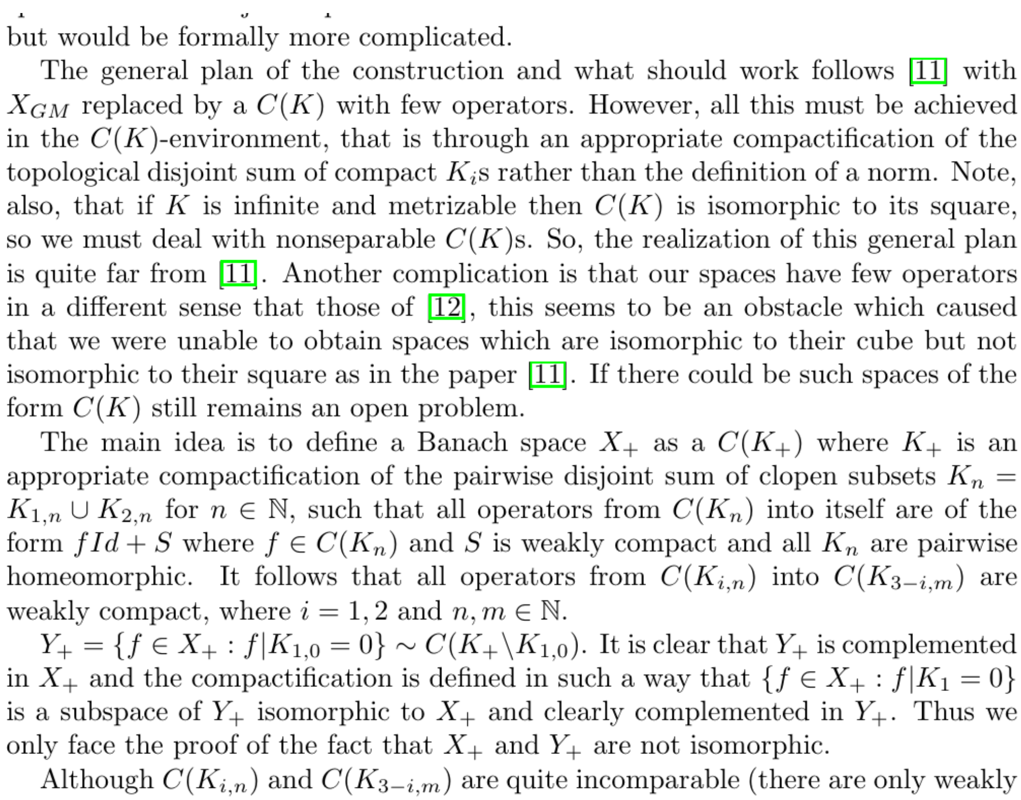
Source Problem

The source paper is Piotr Koszmider, *A $C(K)$ Banach space which does not have the Schroeder-Bernstein property*, arXiv:1106.2917. In the introduction, after explaining why the Gowers-Maurey cube-not-square construction is difficult to reproduce in the $C(K)$ setting, Koszmider asks whether such spaces can be of the form $C(K)$.



but would be formally more complicated.

The general plan of the construction and what should work follows [11] with X_{GM} replaced by a $C(K)$ with few operators. However, all this must be achieved in the $C(K)$-environment, that is through an appropriate compactification of the topological disjoint sum of compact K_is rather than the definition of a norm. Note, also, that if K is infinite and metrizable then $C(K)$ is isomorphic to its square, so we must deal with nonseparable $C(K)$s. So, the realization of this general plan is quite far from [11]. Another complication is that our spaces have few operators in a different sense that those of [12], this seems to be an obstacle which caused that we were unable to obtain spaces which are isomorphic to their cube but not isomorphic to their square as in the paper [11]. If there could be such spaces of the form $C(K)$ still remains an open problem.

The main idea is to define a Banach space X_+ as a $C(K_+)$ where K_+ is an appropriate compactification of the pairwise disjoint sum of clopen subsets $K_n = K_{1,n} \cup K_{2,n}$ for $n \in \mathbb{N}$, such that all operators from $C(K_n)$ into itself are of the form $fId + S$ where $f \in C(K_n)$ and S is weakly compact and all K_n are pairwise homeomorphic. It follows that all operators from $C(K_{i,n})$ into $C(K_{3-i,m})$ are weakly compact, where $i = 1, 2$ and $n, m \in \mathbb{N}$.

$Y_+ = \{f \in X_+ : f|K_{1,0} = 0\} \sim C(K_+ \backslash K_{1,0})$. It is clear that Y_+ is complemented in X_+ and the compactification is defined in such a way that $\{f \in X_+ : f|K_1 = 0\}$ is a subspace of Y_+ isomorphic to X_+ and clearly complemented in Y_+. Thus we only face the proof of the fact that X_+ and Y_+ are not isomorphic.

Although $C(K_{i,n})$ and $C(K_{3-i,m})$ are quite incomparable (there are only weakly

Figure 1: The source open problem on page 2 of arXiv:1106.2917, together with the beginning of the countable compactification construction.

The relevant construction in the source defines Boolean algebras A_n and subalgebras C_n, all copied from a fixed pair (A, C), and then defines

$$D_+ = \left\{ a \in \prod_{n \in \mathbb{N}} A_n : a_n \in C_n \text{ for all but finitely many } n \right\}.$$

The compact space K_+ is the Stone space of D_+, and $X_+ = C(K_+)$.

Result

We prove that this countable-tail template is automatically isometric to all of its finite powers. Consequently, Koszmider's actual K_+ construction, and any direct finite-cycle variant with the same countably repeated tail condition, cannot produce a $C(K)$ space isomorphic to its cube but not to its square.

Proof Intuition

The tail condition "all but finitely many coordinates lie in C_n" is insensitive to splitting $\mathbb{N}$ into finitely many infinite subsequences. A sequence is eventually in the C_n's on all of $\mathbb{N}$ if and only if its restrictions are eventually in the C_n's on each piece of the partition. Since Koszmider's pairs (A_n, C_n) are copies of one fixed pair, each infinite subsequence gives the same Boolean algebra again after reindexing. Stone duality then turns a Boolean product into a topological disjoint union, and C of a finite disjoint union is the finite ℓ_∞-sum.

Theorem 1 (countable-tail finite self-similarity)**.** *Let I be a countably infinite set. For each $i \in I$, let C_i be a Boolean subalgebra of a Boolean algebra A_i, and define*

$$D(I) = \left\{ (a_i)_{i \in I} \in \prod_{i \in I} A_i : a_i \in C_i \text{ for all but finitely many } i \right\}.$$

Let $I = I_1 \sqcup \cdots \sqcup I_r$ be a finite partition into infinite sets. Then restriction to the pieces gives a Boolean algebra isomorphism

$$D(I) \cong D(I_1) \times \cdots \times D(I_r),$$

where each $D(I_j)$ is defined by the same tail rule on I_j.

If, in addition, each subsystem $(A_i, C_i)_{i \in I_j}$ is pair-isomorphic to $(A_i, C_i)_{i \in I}$, then $D(I_j) \cong D(I)$ for every j, and therefore

$$\mathrm{Stone}(D(I)) \cong \bigsqcup_{j=1}^{r} \mathrm{Stone}(D(I)).$$

Hence

$$C(\mathrm{Stone}(D(I))) \cong \underbrace{C(\mathrm{Stone}(D(I))) \oplus_\infty \cdots \oplus_\infty C(\mathrm{Stone}(D(I)))}_{r \text{ copies}}$$

isometrically.

Proof. For $a = (a_i)_{i \in I} \in D(I)$, let $\rho_j(a) = (a_i)_{i \in I_j}$. The map

$$\rho : D(I) \to D(I_1) \times \cdots \times D(I_r), \qquad \rho(a) = (\rho_1(a), \ldots, \rho_r(a)),$$

is a Boolean homomorphism, because all Boolean operations are coordinatewise.

It is injective since the restrictions to the partition pieces determine every coordinate of a. It is surjective as follows. Given $b^{(j)} \in D(I_j)$ for $j = 1, \ldots, r$, define $a_i = b_i^{(j)}$ when $i \in I_j$. For each j, the exceptional set

$$F_j = \{ i \in I_j : b_i^{(j)} \notin C_i \}$$

is finite. Since there are only finitely many pieces, $F_1 \cup \cdots \cup F_r$ is finite, so $a \in D(I)$. Thus ρ is an isomorphism.

Under the extra pair-isomorphism hypothesis, each $D(I_j)$ is simply a reindexed copy of $D(I)$. This gives $D(I) \cong D(I)^r$ as a finite Boolean product. Stone duality sends a finite Boolean product to the topological disjoint sum of the corresponding Stone spaces. Finally, for a finite disjoint union $K = K_1 \sqcup \cdots \sqcup K_r$,

$$C(K) \cong C(K_1) \oplus_\infty \cdots \oplus_\infty C(K_r)$$

isometrically by restriction to the clopen summands. □

Corollary 1 (Koszmider's K_+)**.** *For the compact space K_+ constructed in arXiv:1106.2917,*

$$C(K_+) \cong C(K_+)^r$$

isometrically for every finite $r \geq 1$.

Proof. In Koszmider's construction, the algebras A_n are copies of one Boolean algebra A, and the C_n are the corresponding copies of one subalgebra C. Thus every infinite subsequence of the block indices gives a reindexed copy of the same pair system. Apply the theorem to any partition of $\mathbb{N}$ into r infinite subsets. □

Corollary 2 (obstruction to the direct finite-cycle route)**.** *No construction whose compactifying Boolean algebra has the above countably repeated tail form can witness the cube-not-square phenomenon for $C(K)$.*

Proof. The theorem gives $C(K) \cong C(K)^2$ isometrically as soon as it gives any finite self-similarity at all. Hence such a space cannot satisfy $C(K) \cong C(K)^3$ while $C(K) \not\cong C(K)^2$. □

Limitations

This is a partial result only. It does not rule out a different compact space K, a non-countable-index compactification, or a construction whose finite-power behavior is controlled by a genuinely nontrivial type monoid rather than by a countable eventually-in-subalgebra tail condition. The global question whether some $C(K)$ is isomorphic to its cube but not its square remains open here.

Novelty and Verification Notes

The proof has no computational dependency. It is an explicit Boolean-algebra isomorphism followed by standard Stone duality and the elementary identification of C of a finite disjoint union with a finite ℓ_∞ direct sum.

Bounded local source search for the exact phrases "isomorphic to its cube", "not isomorphic to its square", and close variants found the known Gowers-Maurey phenomenon, several square-negative $C(K)$ or $C_0(K)$ examples, and the prior no-promotion attempt for arXiv:1106.2917. It did not find a later local source resolving Koszmider's global $C(K)$ cube-not-square problem.

Human-review recommendation: check that the intended scope is recorded narrowly. The packet claims a route obstruction for countably repeated tail compactifications, not a full negative answer to Koszmider's open problem.

References

[1] P. Koszmider, *A $C(K)$ Banach space which does not have the Schroeder-Bernstein property*, arXiv:1106.2917.